Nicolò **Favagrossa**

Profile Summary

Physics Master's student specializing in Particles and Astroparticles with 2+ years of R&D experience and active involvement in the JUNO international collaboration. Proven expertise in developing C++ (CERN ROOT) and Python data analysis tools for detector characterization and cosmogenic background reduction. Having worked in both fast-paced startups and large-scale collaborations, I am highly motivated to provide dedicated technical support to CERN's large-scale projects.

Contacts

@ nicolo.favagrossa@gmail.com
+39 389 0155686
Nicolò Favagrossa

Personal Information

Citizenship: **Italian**

Languages:
Italian (Native)
English C1 (University Certified)

Soft Skills

- Scientific Communication
- Logico-Quantitative Analysis
- Data Management & Interpretation
- High-Pressure Problem Solving
- Autonomous Decision Making

Professional Experience

Junior R&D Lab Technician | *Prometheus S.p.A. (Promoted from Assistant R&D Technician in Jan 2025)* **Nov 2023 – Present**

◇ Joined an early-stage energy startup, supporting its transition from a small laboratory to a large-scale R&D facility at Kilometro Rosso.
◇ Proposed and implemented original experimental campaigns and data acquisition strategies based on theoretical insights.
◇ Developed numerical models using Python and C++ to validate physical phenomena; performed extensive scientific literature reviews to redesign experimental setups; calibrated complex sensors for high-precision measurement campaigns.
◇ Acted as a technical bridge between internal R&D, academic partners, and investors, delivering high-level scientific reports.
◇ Collaborated within a heterogeneous and multidisciplinary team, assisting Senior Nuclear Engineers in the setup and operation of specialized nuclear instrumentation; developed practical hardware problem-solving skills in high-pressure laboratory environments.

Education

Master's Degree in Physics
Università degli Studi di Milano **2025 – Present**
◇ Specialization: Particle and astroparticle Physics.

Bachelor's Degree in Physics
Università degli Studi di Milano **2020 – 2024**
Final Grade: 105/110 (GPA details included in academic transcript).
◇ Thesis Title: Cosmic Muons in JUNO: Data Analysis of the Detector Filling Phase

Research Experience

Undergraduate Researcher (Bachelor Thesis) | *JUNO International Collaboration* **2024**

◇ Characterized the cosmic muon flux for the Jiangmen Underground Neutrino Observatory (JUNO) during the finale detector filling phase.
◇ Developed an independent C++ (CERN ROOT) data analysis method to identify cosmic muon events from raw data; performed comparative benchmarking against official JUNO collaboration algorithms to validate event selection efficiency and detector response.
◇ Successfully measured muon rates (compatible with the Monte Carlo expectation) and contributing to the optimization of detector veto strategies, essential for future solar neutrino measurements and cosmogenic background reduction.

Technical and Others Skills

◇ **Programming:** C++ (CERN ROOT, Simulations), Python (Modelling, Data Analysis), Linux (Bash, Terminal), Git/GitHub.
◇ **Tools:** LaTeX (Scientific Documentation), Microsoft Office (Advanced Excel), Data Acquisition Software.
◇ **Laboratory:** Nuclear Instrumentation, Sensor Calibration, Proactive Bibliographic Research.